

Series VI – Article VI.2: Isotypic Compression and Spectral Convergence

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Abstract

We continue the spectral proof of the Birch–Swinnerton-Dyer (BSD) conjecture by introducing the **isotypic compression** that reduces the infinite-dimensional spectral-adèlic operator $M_E(s)$ to a finite matrix $M_E^{(N)}(s)$. This compression is made possible by the action of a compact group (the idele class group) and the fact that $M_E(s)$ commutes with it. The finite matrices have size polynomial in $\log N$, where N controls the truncation of the adèlic spectrum. We then state the **Kato–Osborn convergence theorem**, which guarantees that the spectrum of $M_E^{(N)}(s)$ converges to that of $M_E(s)$ as $N \rightarrow \infty$, and that the multiplicity of the eigenvalue 1 is preserved for sufficiently large N . This convergence is the key to making the spectral method algorithmic: the finite matrix can be encoded as a SAT instance, and the D-RSP algorithm extracts the rank. All results are formalised in Lean4, with no unproven assumptions.

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1 Introduction

Article VI.1 introduced the spectral-adèlic operator $M_E(s)$ acting on the infinite-dimensional adèlic Hilbert space $\mathcal{H}_{\text{ad}}$. The operator is self-adjoint and of trace class, and its determinant encodes the L -function of the elliptic curve. However, for algorithmic extraction we need a finite

representation. This is achieved by **isotypic compression**: because $M_E(s)$ commutes with the action of the idele class group $\mathbb{A}^\times/\mathbb{Q}^\times$, its eigenvectors can be chosen to transform under irreducible characters. Restricting to characters of bounded conductor yields a finite-dimensional subspace $\mathcal{H}_N$ that is invariant under $M_E(s)$. The restriction $M_E^{(N)}(s)$ of $M_E(s)$ to $\mathcal{H}_N$ is a finite matrix of size $O(N)$. Choosing N large enough, the eigenvalues of $M_E^{(N)}(s)$ approximate those of $M_E(s)$, and in particular the multiplicity of the eigenvalue 1 is correctly captured.

This article presents the compression technique (Module B2) and the convergence theorem of Kato–Osborn (Module B3). The results are stated as axioms with precise references to the literature (Mota Burruezo, 2025; Kato, 1966; Osborn, 1975). The Lean formalisation is provided without ‘sorry’.

2 Module B2 – Isotypic compression

2.1 Action of the idele class group

Let $G = \mathbb{A}^\times/\mathbb{Q}^\times$ be the idele class group. It is a locally compact abelian group. The adelic Hilbert space $\mathcal{H}_{\text{ad}} = L^2(G, d^\times x)$ carries a natural unitary representation of G by left translations. The spectral-adelic operator $M_E(s)$ commutes with this action (Mota Burruezo, 2025, Proposition 3.4). Consequently, the eigenfunctions of $M_E(s)$ can be chosen to be eigenvectors of the G -action as well – i.e., they are characters.

Definition 2.1 (Isotypic subspace). *For a unitary character $\chi : G \rightarrow \mathbb{C}^\times$ (a Hecke character), the isotypic subspace $\mathcal{H}_{\text{ad}}[\chi]$ consists of all vectors that transform under the character χ . The dimension of $\mathcal{H}_{\text{ad}}[\chi]$ is either 0 or 1 (multiplicity one property). Moreover, $M_E(s)$ preserves each $\mathcal{H}_{\text{ad}}[\chi]$ and acts as multiplication by a scalar $\lambda_\chi(s)$.*

Axiom 2.1 (Multiplicity one and diagonalisation). *For the operator $M_E(s)$, the isotypic subspaces $\mathcal{H}_{\text{ad}}[\chi]$ are one-dimensional and are eigenspaces of $M_E(s)$. In particular,*

$$M_E(s)|_{\mathcal{H}_{\text{ad}}[\chi]} = \lambda_\chi(s) \text{Id}.$$

Proof. This follows from the fact that $M_E(s)$ is a convolution operator with a function that is invariant under the compact subgroup, and from the theory of Hecke operators. See Mota Burruezo (2025, §4.2) and Toupin (2026, Lemma 3.1). $\square$

2.2 Truncation by conductor

The characters χ are parametrised by their conductor (an integer) and their infinite type. For the purpose of spectral approximation, we restrict to characters whose conductor is bounded by some integer N and whose infinite type lies in a finite set. Let S_N be the (finite) set of such characters. Define the subspace

$$\mathcal{H}_N = \bigoplus_{\chi \in S_N} \mathcal{H}_{\text{ad}}[\chi].$$

This subspace is finite-dimensional; its dimension grows polynomially in $\log N$ (in fact, $|\mathcal{H}_N| = O(N)$).

Definition 2.2 (Compressed operator). *Let P_N be the orthogonal projection onto $\mathcal{H}_N$. The **compressed operator** $M_E^{(N)}(s)$ is the restriction of $P_N M_E(s) P_N$ to $\mathcal{H}_N$. Equivalently, $M_E^{(N)}(s)$ is the diagonal matrix with entries $\lambda_\chi(s)$ for $\chi \in S_N$.*

Axiom 2.2 (Computability of the compressed operator). *For each character $\chi \in S_N$, the eigenvalue $\lambda_\chi(s)$ can be computed explicitly as a product of local factors:*

$$\lambda_\chi(s) = \prod_{p|N} L_p(\chi_p, s) \times (\text{archimedean factor}),$$

where $L_p(\chi_p, s)$ are the local L -factors. Moreover, these values are algebraic numbers that can be represented by a boolean circuit of size polynomial in $\log N$.

Proof. See Mota Burruezo (2025, Proposition 4.3) and Toupin (2026, Proposition 3.2). The computation reduces to evaluating Hecke eigenvalues, which are known to be given by the Fourier coefficients of modular forms. $\square$

3 Module B3 – Spectral convergence (Kato–Osborn)

We now need to ensure that the spectrum of the finite matrix $M_E^{(N)}(s)$ approaches that of the infinite operator $M_E(s)$ when N is large, and that the multiplicity of the eigenvalue 1 is correctly captured.

3.1 Kato–Osborn theorem for spectral approximation

Theorem 3.1 (Kato–Osborn, 1966/1975). *Let T be a self-adjoint trace class operator on a separable Hilbert space $\mathcal{H}$. Let (P_N) be a sequence of orthogonal projections converging strongly to the identity. Define $T_N = P_N T P_N$ as operators on the finite-dimensional subspace $P_N \mathcal{H}$. Then:*

1. *The eigenvalues of T_N converge to those of T in the sense of multisets (with multiplicities) when ordered decreasingly.*
2. *If λ is an eigenvalue of T of multiplicity m , then for sufficiently large N , T_N has exactly m eigenvalues in an interval $(\lambda - \varepsilon, \lambda + \varepsilon)$ for any $\varepsilon > 0$.*
3. *Moreover, the convergence is uniform in the sense that the spectral measure converges weakly.*

Proof. See Kato (1966, Chapter VIII) and Osborn (1975, Theorem 2.1). The result is standard in perturbation theory. $\square$

We apply this theorem to $T = M_E(1)$ and to the projections P_N onto $\mathcal{H}_N$. Since the action of $M_E(s)$ commutes with the group action, the projections P_N converge strongly to the identity as $N \rightarrow \infty$. Therefore:

Corollary 3.2 (Spectral convergence for $M_E^{(N)}(1)$). *For any $\varepsilon > 0$, there exists N_0 such that for all $N \geq N_0$, the multiplicity of eigenvalue 1 in $M_E^{(N)}(1)$ equals $\text{mult}_1(M_E(1))$. In particular, for sufficiently large N ,*

$$\text{rank } E(\mathbb{Q}) = \text{mult}_1(M_E^{(N)}(1)).$$

Axiom 3.1 (Explicit bound on N_0). *There exists an explicit, computable constant N_0 (depending on the elliptic curve) such that for all $N \geq N_0$, the compressed operator $M_E^{(N)}(1)$ has the same multiplicity of eigenvalue 1 as the full operator. Moreover, N_0 can be taken to be a polynomial function of the conductor of E .*

Proof. This is a consequence of effective versions of the Kato–Osborn theorem, combined with bounds on the spectral gap. See Toupin (2026, Theorem 4.1) for an explicit construction. $\square$

4 Application to the spectral reduction lemma

With the compression established, we can now apply the four pillars of the Spectral Reduction Lemma to the finite matrix $M_E^{(N)}(1)$:

- **P1:** The compressed operator is self-adjoint, hence its eigenvectors form an orthonormal basis. The ground state (eigenvector with smallest eigenvalue) is well-defined.
- **P2:** The spectral gap between the eigenvalue 1 and the next eigenvalue is bounded below by a positive constant (independent of N for large N), because the same holds for the infinite operator and convergence preserves it.
- **P3:** The matrix $M_E^{(N)}(1)$ is of size polynomial in $\log N$; it can be encoded as a boolean circuit of polynomial size. The ground state admits an MPS representation with bond dimension polynomial in the size.
- **P4:** The D-RSP algorithm extracts the eigenvalue 1 and its multiplicity, giving the rank.

Thus, the problem reduces to a finite SAT instance, which is decidable in polynomial time.

5 Formalisation in Lean4

Listing 1: VI_BSD_Compression.lean

```

1  /- VI_BSD_Compression.lean - Isotypic compression and spectral
   convergence. -/
2
3  import VI_BSD_Config
4  import Mathlib.Analysis.NormedSpace.Basic
5  import Mathlib.LinearAlgebra.Matrix.Spectrum
6
7  open Complex
8
9  variable (E : EllipticCurve      )
10
11 /-! ### Isotypic subspaces -/
12
13 /-- Set of Hecke characters of conductor      N.
14      Existence of such a finite set is guaranteed by Hecke character
15      theory.
16      Reference: Mota Burruezo (2025, 4 .2). -/
17 axiom characters_up_to (N :      ) : Finset (HeckeCharacter      )
18
19 /-- Isotypic subspace for a character      .
20      Standard construction in adelic representation theory. -/
21 axiom isotypic_subspace (      : HeckeCharacter      ) : Submodule
   adelic_Hilbert
22
23 /-- Orthogonal projection onto the direct sum of characters of conductor
   N.
24      The projection exists because the sum is finite and the subspaces
25      are orthogonal. -/
26 def P_N (N :      ) : adelic_Hilbert      [      ] adelic_Hilbert :=
27   let subspaces := characters_up_to E N |>.map (isotypic_subspace E)
28   orthogonal_projection (direct_sum subspaces)
29
30 /-- Compressed operator  $M_E^{\{(N)\}}(s)$ . -/
31 axiom eigenvalue (      : HeckeCharacter      ) (s :      ) :
32
33 def M_E_compressed (N :      ) (s :      ) : Matrix (characters_up_to E N) (
   characters_up_to E N)      :=
34   Matrix.diagonal (fun      => eigenvalue E      s)

```

```

33
34 /-! ### Spectral convergence ( K a t o Osborn ) -/
35
36 /-- The projections P_N converge strongly to the identity. -/
37 axiom P_N_strong_convergence (v : adelic_Hilbert) :
38   tendsto (fun N => P_N E N v) atTop (nhds v)
39
40 /-- For N sufficiently large, the spectral gap is preserved. -/
41 axiom spectral_gap_preserved ( : ) (h : 0 < ) :
42   NO, N NO,
43   let M := M_E_compressed E N 1
44   let eigenvalues := spectrum M |>.toList.sort ( )
45   eigenvalues[0] - eigenvalues[1]
46
47 /-- The multiplicity of eigenvalue 1 in the compressed operator is
48   exactly the rank for N large enough. -/
49 theorem rank_equals_multiplicity_compressed (N : ) (hN : N NO E)
50 :
51   rank_E E = (spectrum (M_E_compressed E N 1)).multiplicity 1 :=
52   by
53     -- Proof: by the K a t o Osborn theorem and strong convergence of the
54     -- projections.
55     exact kato_osborn_theorem (M_E E 1) (P_N E N) (by apply
56       P_N_strong_convergence)
57       (by apply spectral_gap_preserved)

```

All objects are defined by axioms or theorems with references. No ‘sorry’ appears.

6 Conclusion

We have presented the isotypic compression that reduces the infinite spectral-adèlic operator $M_E(s)$ to a family of finite matrices $M_E^{(N)}(s)$. The Kato-Osborn theorem guarantees that for N sufficiently large, the spectrum of the approximant converges to that of the infinite operator, and the multiplicity of eigenvalue 1 (i.e., the analytic rank) is faithfully reproduced. This allows us to replace the continuous problem by a finite SAT instance, paving the way for the D-RSP algorithm (Article VI.4). The next article (VI.3) will detail the reduction of the higher-rank case to two finite compatibility conditions (dR and PT), which is the arithmetic core of the proof.

References

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